

ACMS 40876/60876 (Spring 2026): Final Project

Final Project Instructions

The final project for this course is fairly open-ended (be creative!) but should demonstrate a mastery of the full data science life cycle **in practice** (either through the tools used or its applications). In addition to this “data science in practice” component, two other crucial skills of any effective data scientist involve coding practices and clear communication. As such, this final project will consist of (1) a coding component, (2) written report, and (3) short oral presentation. Details regarding each of these three components and their grading can be found in subsequent sections.

Moreover, there will be three checkpoints in order to provide opportunities for intermediary feedback on the final project. These checkpoints will be graded based upon completion and are used primarily to exchange feedback. All due dates are at **11:59pm ET**.

Due Dates.

Checkpoint 1. Project proposal due **Friday, March 6**

Checkpoint 2. First code review due **Friday, April 3**

Checkpoint 3. Report draft and second code review due **Friday, April 17**

Final presentation. During last two class periods (**April 23 and 28**)

Final report + code. Due **Thursday, May 7**

Final Project Checkpoint 1

In the first final project checkpoint, please write a brief project proposal (maximum 1 page) describing your plans for the final project. The proposal must include responses to the following questions:

1. What is the high-level problem that you'd like to explore in this final project?
2. Why is this problem interesting, or why did you choose this problem?
3. How does this problem relate to the course or the data science life cycle?
4. What are the high-level steps you plan to take to tackle this problem?
5. If applicable, provide a link to the data source that you intend to use.
6. (*ACMS 40876 Only*) List your partner's name if you are working with a partner.

Submit your proposal to your GitHub at `dsip/final_project/documents/proposal.pdf`.

Final Project Checkpoint 2

The second final project checkpoint will be a code review.

1. Push all of the code that you have written so far to GitHub for your first code review. This code should be written in your `final_project/` folder and organized using a file structure similar to your labs.

Tip: If you are completing a real data analysis project, it is recommended to have conducted a preliminary exploratory data analysis and modeling pass through the data. This would allow for more constructive feedback at this checkpoint.

Details on how to perform this code review (using [GitHub flow](#)) will be discussed during class when it gets closer.

Final Project Checkpoint 3

The third final project checkpoint includes (1) a second code review and (2) an outline or draft of the final project report.

1. Please address the issues from the first code review, following the usual [GitHub flow](#). Push these revisions as well as all other code that you have written to GitHub for your second code review.
2. Submit a partial or full draft of your final project report by pushing it to your GitHub repository at `dsip/final_project/documents/report_draft.pdf`. While the draft need not be complete, you are highly encouraged to write as much of the report as you can in order to obtain better feedback. At a minimum, a detailed outline with tentative section headers/subheaders and placeholders for your intended figures is required.

1 Final Project: Code

Your final project coding component grade will be based upon:

- Use of GitHub flow to address feedback from checkpoints
- Reproducibility: all results/figures should be easily reproduced with a single click of a button using reproducible documentation tools (e.g., R Markdown, quarto, and/or Jupyter notebooks) and reproducible environments (e.g., `renv` or `conda`)
- Overall code organization and style
 - When writing code, please adhere to a consistent coding style. Some recommendations include following the [tidyverse style guide](#) in R and [PEP 8](#) in Python.

Rubric

- **Code Reproducibility** (15 points total)
 - Are all packages/dependencies included and easily installable in a reproducible environment (using `renv`, `conda`)? (10 points)
 - Can all results be easily reproduced (e.g., by following the steps outlined in the README and/or by rendering a single notebook)? (5 points)
 - * Note: large data files should not be pushed to GitHub; instead, please provide a link to where the data can be downloaded and detail how to download this data in your Readme
- **Code Style** (15 points total)
 - Are functions being created and documented appropriately? In other words, is the code modular? (5 points)
 - Is the coding style consistent? E.g., for R users, was the tidyverse style guide followed, and for Python users, was the PEP-8 style guide followed? (5 points)
 - Is the `final_project` file directory appropriately structured and organized (as was done in all of the previous labs)? (2 points)
 - Are there comments throughout the code? (3 points)
- **GitHub Flow:** (10 points total)
 - Was GitHub flow followed during the code review process?

2 Final Project: Written Report

Submit your final project report to GitHub at `final_project/documents/final_report.pdf` by **May 7 at 11:59pm ET**. Due to end-of-semester grading deadlines, I cannot grant any deadline extension requests.

Given the open-ended nature of this final project, the structure of each student's final report can vary. However, a sample checklist for the final project report is provided below. Note that not all checklist items are applicable to your final project, and the grading rubric will be adjusted accordingly.

Sample Checklist

- **Introduction (Problem Formulation).** The introduction should, at a minimum, address each of the following:
 - What is the big-picture scientific problem that you are interested in?
 - Why is this scientific problem interesting or relevant to researchers in the field and/or general members of society?
 - Are there particular challenges that make solving this problem difficult? Here, you may want to briefly discuss prior work and their limitations
 - What is the specific aim or contribution of this present work, as it relates to the big-picture scientific problem?
- **Data Collection.** An overview of the data and the data collection process should be described, addressing:
 - How was the data collected or generated?
 - * Note: a general reader with no knowledge of the original study should be able to understand how the data was collected at a high-level by reading only your data description. This description should also provide enough information to understand the remainder of your analysis.
 - Briefly describe the variables in the data, including how these variables were measured/collected and why they are of interest (when possible). Please also make a concerted effort to justify why at least one of the variables or the set of variables as a whole are relevant to this study.
 - Are there any special properties of the data/data collection that make it uniquely suited to answer your scientific question of interest? At a minimum, be sure to relate the data to your scientific problem.
 - Are there any limitations when using the data to answer the domain problem of interest? Be as transparent as possible so that conclusions made from this data are not misinterpreted down the road.

- **Data Preprocessing.** Please include the following regarding your data preprocessing and cleaning steps:
 - Discuss all inconsistencies, problems, and oddities that you find with the data. These may include, but are not limited to: missing values, erroneous measurements, and outliers.
 - Record the steps that you took to clean the data. These steps should be written in sufficient detail such that the reader can exactly reproduce your data cleaning procedure based upon your description.
 - When possible, explain why you cleaned the data in that way when possible. Here, you may choose to bring in domain knowledge or common knowledge to support your data cleaning decisions.
 - Please also include any exploratory plots you created to help identify these data issues and/or guide your data preprocessing choices.
- **Methods.** Descriptions of the methodology and analysis pipeline should be provided, including:
 - How was data splitting performed? Why was the data splitting performed in this way, or why was this scheme appropriate for your particular problem?
 - What methods/models were used in your analysis pipeline and why were these methods chosen? Your justification should convey that these modeling choices were not arbitrary choices, but rather the result of critical thinking.
 - How did you evaluate your methods? Why did you choose these evaluation metrics? Note that a good evaluation typically involves multiple evaluation metrics.
 - Were additional post-hoc analyses performed? This can include investigations surrounding interpretability, local error diagnostics, and other interesting avenues that arise from your initial modeling.
 - Be sure to include enough detail that your modeling steps can be exactly reproduced based upon your writeup. (This includes detailing your model hyperparameters, if applicable.)
- **Results.** Results can include both exploratory data analyses and findings from your more formal modeling. Grading of this section is likely dependent on the particular final project and is thus purposefully very broad. Generally, you will be graded based upon:
 - **Content quality:** Has the domain problem been meaningfully addressed by the performed analysis and the resulting findings?
 - **Critical discussion and relevance:** Are the findings discussed in a critical and thoughtful manner? Moreover, are the findings relevant and well-connected to the motivating domain problem?

- **Accuracy:** Are the findings conveyed truthfully and accurately (not misrepresented)?
- **Discussion.** Please conclude your final report with a short discussion section, including a summary of your analysis and findings in the context of the broader domain. You may also want to include a brief discussion of limitations or future work, as if this were a research manuscript.
- **Overall Quality**
 - **Figures:** Are the figures publishable quality? Be sure to:
 - * Add captions that include the main takeaway of the figure
 - * Format the axes and legend text/titles appropriately
 - * Choose visually appealing colors that aid readability of the figure
 - * Choose appropriate levels of transparency and sizes of points, lines, etc.
 - * Choose an appropriate plot type to convey your message
 - **Narrative:** Is the overall narrative cohesive (with meaningful transitions between sections) and has a clear and focused takeaway?
 - **Readability/Grammar:** Is there an excellent command of grammar throughout the report?

3 Final Project: Presentation

Final project presentations will take place during class on **April 23** and **28**.

Rubric. Your final project presentation will be graded based on the following rubric.

Oral Presentation Rubric				
Criteria	Excellent (5)	Good (4)	Fair (3)	Needs Improvement (0-2)
Content and Analysis	Meaningful analysis and full knowledge of the domain and statistical topics are demonstrated. Questions from the audience are answered critically.	Appropriate analysis and reasonable knowledge of the domain and statistical topics are demonstrated with only minor gaps in knowledge. Questions from the audience are answered reasonably.	The content and analysis shows some understanding but has noticeable gaps regarding either the domain or statistical topics under study.	The content and analysis shows severe flaws in the domain and/or statistical topics under study.
Organization and Structure	Presentation is organized in a logical and effective way that can be easily understood. Timing guidelines are followed.	Presentation is organized in a logical way that can be understood within reason. Timing guidelines are followed.	Presentation is somewhat difficult to follow, but the general idea can be understood.	Presentation is difficult to follow and lacks logical organization.
Visuals	Visuals and slides are thoughtfully and creatively constructed with excellent aesthetics. Visuals distill ideas in a very clear and accurate manner.	Visuals and slides are aesthetically-pleasing and help to facilitate understanding, but leave room for minor improvements.	Visuals and slides are mildly pleasing and somewhat facilitate understanding, but leave room for major improvements.	Visuals and slides are difficult to parse and require a major overhaul.
Communication and Oral Delivery	Presentation is delivered in a very effective and engaging manner. Maintains eye contact, doesn't read from notes, speaks loud with inflection, etc.	Presentation delivery is adequately effective and engaging.	Presentation delivery is mildly effective and engaging.	Presentation delivery is not effective nor engaging. Does not make eye contact, reads from notes, speaks quietly.